

Chapter 10

AI-Driven Solutions for Social Impact Transforming Businesses for a Better Future

Ranjan Banerjee

 <https://orcid.org/0009-0003-1950-7530>

Brainware University, India

Malay Maity

Brainware University, India

Ananya Das

*Supreme Institute of Management and
Technology, India*

Sulekha Das

Brainware University, India

Pranab Gharai

Brainware University, India

Ishita Mondal

Swami Vivekananda University, India

Subhadip Sarkar

Techno India, India

Sudeshna Ghosh

Techno India, India

ABSTRACT

Financial exclusion remains a significant barrier to poverty reduction and economic development. Social entrepreneurs play a crucial role in bridging this gap, but their reach is often limited. This paper explores the potential of Artificial Intelligence (AI) as a force for financial democracy, empowering social entrepreneurs to extend financial inclusion to underserved communities. We examine how AI-powered solutions like alternative credit scoring, automated loan management, and personalized financial products can unlock financial access for the “last mile.” The paper acknowledges the challenges associated with AI adoption, such as data privacy and algorithmic bias. We conclude by highlighting the need for responsible development and implementation of AI solutions to ensure a truly inclusive and equitable financial system.

DOI: 10.4018/979-8-3693-6392-8.ch010

SETTING THE SCENE

Social entrepreneurs working at the grassroots level are the backbone of sustainable economic development. They address critical social issues like poverty, unemployment, and lack of access to basic necessities through innovative and scalable solutions. However, their impact is often limited by resource constraints and the sheer volume of beneficiaries they cater to. Artificial intelligence (AI) has the potential to revolutionize this landscape by automating tasks, personalizing financial services, and facilitating data-driven decision making for microfinance institutions (MFIs) and social enterprises (Kaggwa et. AI.,2024).

This chapter explores the transformative potential of AI in promoting inclusive microfinance and fostering financial inclusion for low-income communities. We delve into specific AI applications that can empower social entrepreneurs, enhance financial services delivery, and ultimately contribute to sustainable economic development.

AI-Powered Credit Scoring: Unlocking Financial Inclusion for the Unbanked

For millions of people around the world, the traditional credit scoring system presents a significant barrier. Those without a formal banking history or a credit card often lack the creditworthiness data required to access loans and other financial products. This exclusion, often referred to as being “unbanked,” can perpetuate a cycle of poverty and limit opportunities for financial growth. However, Artificial Intelligence (AI) is emerging as a powerful tool to bridge this gap and unlock financial inclusion for the unbanked population.

Challenges of Traditional Credit Scoring

Traditional credit scoring models rely heavily on factors like credit card history, loan repayment records, and income verification. For individuals who haven't participated in the formal financial system, this data is simply unavailable, making them invisible to traditional lenders. They often rely on credit history and financial data, which a significant portion of the unbanked population lacks (Bialkova et. AI.,2024).AI-powered alternative credit scoring can leverage alternative data sources like mobile phone usage patterns, utility bill payments, and social network activity to assess creditworthiness. This enables MFIs to extend financial services to previously excluded populations, fostering financial inclusion and economic empowerment.

AI-Driven Loan Collection and Risk Management

Manual loan collection processes can be time-consuming and resource-intensive for MFIs. AI-powered chatbots can automate initial communication with borrowers, provide reminders, and answer basic queries. Additionally, AI algorithms can analyze repayment patterns and identify potential defaulters early on, allowing MFIs to take proactive measures and improve loan collection rates. This frees up resources for MFIs to focus on client outreach and financial literacy initiatives (Aldoseri et. AI.,2024).

18 more pages are available in the full version of this document, which may be purchased using the "Add to Cart" button on the product's webpage:

www.igi-global.com/chapter/ai-driven-solutions-for-social-impact-transforming-businesses-for-a-better-future/366888?camid=4v1

Related Content

Explainable AI-Based Semantic Object Detection for Autonomous Vehicles

Elamathiyan A. and G. Dhivya (2024). *Modeling, Simulation, and Control of AI Robotics and Autonomous Systems* (pp. 15-32).

www.igi-global.com/chapter/explainable-ai-based-semantic-object-detection-for-autonomous-vehicles/348032?camid=4v1a

A Hybrid Portfolio Selection Model: Multi-Criteria Approach in the Indian Stock Market

Praveen Ranjan Srivastava and Prajwal Eachempati (2020). *International Journal of Intelligent Information Technologies* (pp. 100-116).

www.igi-global.com/article/a-hybrid-portfolio-selection-model/257215?camid=4v1a

Dependable Services for Mobile Health Monitoring Systems

Marcello Cinque, Antonio Coronato and Alessandro Testa (2012). *International Journal of Ambient Computing and Intelligence* (pp. 1-15).

www.igi-global.com/article/dependable-services-mobile-health-monitoring/64187?camid=4v1a

Multimodal Human Aerobotic Interaction

Ayodeji Opeyemi Abioye, Stephen D. Prior, Glyn T. Thomas, Peter Saddington and Sarvapali D. Ramchurn (2017). *Smart Technology Applications in Business Environments* (pp. 39-62).

www.igi-global.com/chapter/multimodal-human-aerobotic-interaction/179031?camid=4v1a